

GRAFTON RESEARCH STATION, Australia

WMO: 955710

Lat: 29.6225S

Lon: 152.9606E

Elev: 84

StdP: 14.65

Time Zone: 10.00 (AUS)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 41.2	(c) 43.4	(d) 27.5	(e) 21.5	(f) 60.3	(g) 30.7	(h) 25.0	(i) 58.5	(j) 21.6	(k) 61.8	(l) 18.9	(m) 62.4	(n) 4.6	(o) 290	(p) 0.344

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 17.3	(c) 94.3	(d) 73.1	(e) 90.5	(f) 72.6	(g) 87.5	(h) 71.9	(i) 76.7	(j) 88.9	(k) 75.3	(l) 86.1	(m) 74.0	(n) 83.5	(o) 9.7	(p) 70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.3	124.3	81.2	72.2	119.4	80.2	71.1	114.9	78.9	40.1	89.2	38.8	85.9	37.6	83.7	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 19.4	(b) 16.5	(c) 14.3	(e) 35.6	(f) 102.8	(g) 2.0	(h) 4.4	(i) 34.2	(j) 106.0	(k) 33.0	(l) 108.6	(m) 31.9	(n) 111.1	(o) 30.5	(p) 114.3		
			DB													
			WB													
				33.9	79.2	2.1	1.6	32.4	80.3	31.2	81.3	30.0	82.2	28.5	83.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	67.5	76.6	75.9	73.3	68.4	62.5	58.6	57.4	59.1	64.3	68.2	72.0	74.5
	DBStd	7.81	4.11	3.90	3.41	3.31	3.53	3.52	3.43	3.71	4.19	4.71	4.62	4.52
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	807	0	0	0	8	93	194	238	188	62	20	3	1
	CDD50	6392	824	724	721	551	388	257	228	283	429	565	661	761
	CDD65	1723	359	304	256	109	16	1	1	6	41	120	213	297
	CDH74	11840	2706	2109	1370	472	102	6	13	112	418	936	1500	2096
	CDH80	3936	1041	727	351	62	7	1	0	19	109	311	538	770
(14) Wind	WSAvg	6.1	6.1	6.6	5.9	5.8	5.5	6.0	5.8	6.2	6.2	6.3	6.3	6.3
(15) Precipitation	PrecAvg	42.2	5.1	4.9	5.2	4.5	3.6	2.8	2.2	1.5	1.2	3.0	3.5	4.2
	PrecMax	66.7	11.9	14.1	20.4	16.1	13.9	23.2	11.6	6.3	3.9	22.6	13.0	13.7
	PrecMin	19.5	0.3	0.6	0.1	0.4	0.5	0.2	0.0	0.0	0.0	0.2	0.4	0.9
	PrecStd	11.0	3.1	3.7	3.9	4.4	3.5	4.3	2.7	1.6	1.1	3.9	2.6	2.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	99.7	97.2	92.4	85.0	80.7	74.0	75.9	82.3	88.4	93.4	96.7	100.0
		MCWB	75.3	76.5	74.3	69.6	65.2	60.8	59.9	59.9	64.1	68.0	71.6	72.1
	2%	DB	92.7	91.2	86.7	81.4	76.9	71.2	72.1	76.9	82.7	87.5	90.3	92.0
		MCWB	75.0	75.6	72.9	68.1	63.2	59.7	57.8	58.7	62.5	66.8	70.2	73.0
	5%	DB	88.7	87.4	83.6	78.9	73.9	69.2	69.1	73.1	78.4	82.9	85.9	87.2
		MCWB	74.2	73.2	71.3	66.7	61.9	58.5	56.6	57.2	61.9	65.9	69.6	71.8
	10%	DB	85.6	84.4	81.0	76.5	71.5	67.1	66.6	69.9	75.0	79.0	81.9	83.8
		MCWB	72.7	71.6	69.9	65.7	60.5	57.5	55.6	56.0	61.0	65.0	68.4	70.6
	0.4%	WB	78.6	78.6	76.4	72.0	68.3	64.7	64.7	64.4	68.2	72.3	75.7	77.5
		MCDB	91.7	92.0	87.0	81.3	75.6	69.4	70.7	74.6	80.7	84.3	90.8	90.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	76.6	76.5	74.5	69.9	66.1	62.6	61.7	61.4	66.0	69.8	73.2	75.3
		MCDB	88.7	88.0	83.1	77.7	72.8	68.0	67.1	71.0	77.6	81.3	85.1	87.4
	5%	WB	75.1	74.8	73.1	68.4	64.2	61.0	59.3	59.6	64.4	68.0	71.2	73.6
		MCDB	85.4	84.5	80.5	75.9	70.5	65.9	65.7	68.4	74.5	78.4	81.0	83.9
	10%	WB	73.8	73.2	71.6	67.1	62.5	59.6	57.4	58.1	62.7	66.4	69.7	72.1
		MCDB	82.7	81.8	78.6	74.2	68.9	64.7	63.9	66.6	71.3	75.8	78.7	80.7
(35) Mean Daily Temperature Range	5% DB	MDBR	17.3	16.9	16.2	17.4	19.7	18.5	21.2	23.7	22.8	20.8	19.8	18.1
		MCDBR	23.5	22.3	20.7	20.7	22.7	21.4	24.9	29.3	29.3	28.6	27.0	24.8
		MCWBR	8.8	8.1	8.1	9.1	10.9	11.6	12.7	13.8	12.3	11.6	10.3	9.2
	5% WB	MCDBR	20.8	20.7	18.6	18.2	18.5	16.7	19.6	23.8	24.6	23.3	22.2	21.3
		MCWBR	8.7	8.2	8.0	8.9	9.9	10.1	11.5	12.4	11.7	10.5	9.4	9.0
(40) Clear-Sky Solar Irradiance	taub		0.386	0.373	0.356	0.336	0.310	0.304	0.297	0.311	0.355	0.388	0.385	0.389
	taud		2.505	2.559	2.599	2.627	2.654	2.660	2.662	2.601	2.471	2.413	2.473	2.484
	Ebn at Noon		303	301	297	287	281	276	283	292	292	293	301	303
	Edh at Noon		36	33	30	27	23	22	23	27	34	38	37	37
(44) All-Sky Solar Radiation	RadAvg		2058	1865	1576	1353	1107	922	1058	1347	1683	1872	2004	2055
(45)	RadStd		171	153	99	67	72	57	81	97	123	171	189	194

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (17 neighbors)	+0.63	N/A	N/A	+1.13	N/A	N/A	N/A	-86	+290	+169

Nomenclature: See separate page